

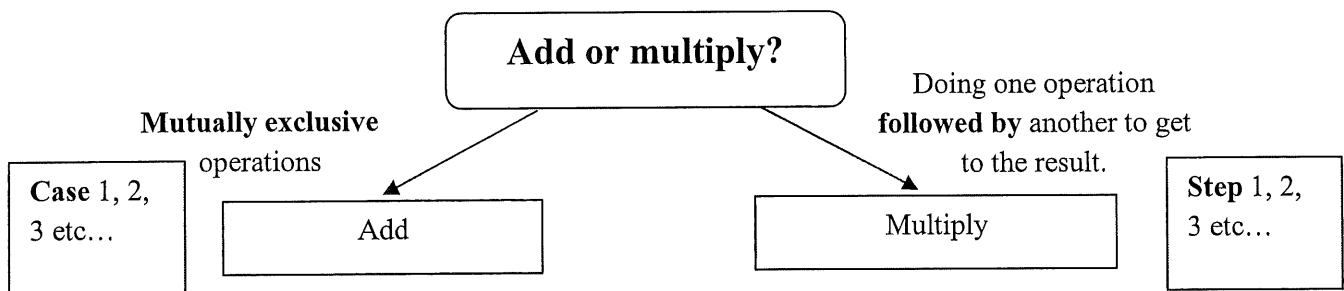


RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 13 (Summary and Tutorial)

Topic: Permutations and Combinations, Probability

Summary for Permutations and Combinations

- Definitions** - A **permutation** is an *ordered arrangement* of objects
 - A **combination** is a *selection of objects* in which the order of selection *does not matter*.



Permutations

Given	Objects taken	No of Permutations
n distinct objects	n	$n!$
n distinct objects	r <i>no repetitions</i>	${}^n P_r$ or ${}^n C_r r!$
n distinct objects	r <i>with repetitions</i>	n^r
n objects, not all distinct (n_1 of type 1, n_2 of type 2, ..., n_k of type k , where $n = n_1 + n_2 + \dots + n_k$)	n	$\frac{n!}{n_1! n_2! \dots n_k!}$
n objects, not all distinct	r	Involves combinations & permutations See section on Combinations .

Useful Techniques when dealing with restrictions:

1. Grouping or Slotting.
 - (a) Grouping ("must be together", "cannot be separated" etc.).
Eg. No of ways to arrange the letters a, b, c, d, e, f, g such that a, b & c are together
 $= (5!) \times (3!)$
 [abc, d, e, f, g: 5 items $\rightarrow$ 5! ways, within abc $\rightarrow$ 3! ways]
 - (b) Slotting ("cannot be together", "must be separated" etc.)

Eg. No of ways to arrange the letters a, b, c, d, e, f, g such that a, b & c are not adjacent to each other $= (4!) \times ({}^5P_3)$
 [arrange d, e, f, g first $\rightarrow 4!$ ways, slot and permute a, b, c $\rightarrow {}^5P_3$ ways]

2. Taking Complement (“two items cannot be together” or “at least 1”).

Use this technique with care. Note that for Eg in 1(b) above, a, b, c not adjacent to each other is NOT the complement of abc together.

Combinations

Notation: nC_r can also be written as $\binom{n}{r}$

Note:

- $\binom{n}{r} = \binom{n}{n-r} = \frac{n!}{r!(n-r)!}$ (Highlighted portion is in MF27). E.g. $\binom{9}{2} = \binom{9}{7}$
- ${}^nP_r = {}^nC_r \cdot r!$

No of ways to select r objects	Given : n <i>distinct</i> objects $\binom{n}{r}$	Given: n objects, not all distinct No direct way to calculate. Need to consider different cases: (i) Start with case where the r selected objects are <i>all distinct</i> . (ii) Next, consider case(s) with pair(s) of identical objects, and so on. (iii) No of ways = Sum of the different cases
Eg: To choose 3 letters (<i>arrangements</i> not required)	Given: Letters a,b,c,d,e,f No of ways $= \binom{6}{3} = 20$	Given: a, a, a, b, b, c Case 1: All distinct (abc) – 1 way. Case 2: Contains an identical pair (aab, aac, bba, bbc) – $2 \times 2 = 4$ ways. Case 3: All identical (aaa) – 1 way. Total no of ways $= 1 + 4 + 1 = 6$
Eg. Find the no of 3-letter codes (<i>arrangements</i> to be considered)	Given: Letters a,b,c,d,e,f No of ways $= \binom{6}{3} (3!) = 120$ OR: ${}^6P_3 = 6 \times 5 \times 4 = 120$	Given: a, a, a, b, b, c Case 1: All distinct (abc) – $1 \times 3! = 6$ ways. Case 2: Contains an identical pair (aab, aac, bba, bbc) – $2 \times 2 \times \frac{3!}{2!} = 12$ ways. Case 3: All identical (aaa) – 1 way. Total no of ways $= 6 + 12 + 1 = 19$

Remark: Before doing any calculations, it is important to establish if

- the problem involves *permutations only*, or *combinations only*, or *both*.
- the objects or items given are *all distinct*. (see Eg in above table).
- repetitions* are allowed.

Circular Permutations – Arranging n distinct objects in a Circle

- In row arrangement, such as in a queue, there is a *first* & a *last* position.

No of ways = $n!$

- In a circle, such as a round table, there is no starting or ending point.

No of ways = $\frac{n!}{n} = (n - 1)!$

- No. of ways to seat 2 couples and a boy at a round table = $(5 - 1)! = 24$
- No. of ways to seat 2 couples and a boy at a round table if each couple is seated together = $(3 - 1)!2!2! = 8$
[couple 1, couple 2, boy round the round $\rightarrow (3 - 1)!$ ways, each couple $\rightarrow 2!$ ways]
- If the seats of the round table are numbered, we will perform the necessary calculations assuming the seats are not numbered, and multiply the value with the number of seats
 - No. of ways to seat 2 couples and a boy at a round table with numbered seats = $(5 - 1)! \times 5 = 120$
 - No. of ways to seat 2 couples and a boy at a round table if each couple is seated together, and the seats are numbered = $(3 - 1)!2!2! \times 5 = 40$

Reminder: Questions on P&C may involve probability. Read the questions carefully.

Summary for Probability

If the sample space S consists of a finite number of **equally likely** outcomes, then the probability of event A written as $P(A)$ is defined as

$$P(A) = \frac{\text{no. of elements in } A}{\text{no. of elements in } S} = \frac{n(A)}{n(S)}.$$

In layman definition, Required probability = $\frac{\text{outcomes we are interested in}}{\text{total possible outcomes}}$.

Important Results

1. $0 \leq P(A) \leq 1$, answers should always be between 0 and 1 inclusive AND exact where possible.
2. $P(A') = 1 - P(A)$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
With the help of Venn diagrams, other results (besides Results 2 and 3) can also be easily deduced. For example,
 $P(A \cup B) = P(A) + P(A' \cap B)$,
 $P(A) = P(A \cap B) + P(A \cap B')$
or other cases like $A' \cup B, A \cup B', A' \cup B', A \cap B, A' \cap B, A \cap B', A' \cap B'$ and even those cases with 3 sets A, B, C .
4. If A and B are **mutually exclusive**, then $P(A \cap B) = P(\phi) = 0$, thus $P(A \cup B) = P(A) + P(B)$.

Conditional Probability

If A and B are two events and $P(B) \neq 0$, then the probability of A , given that B has already occurred, is written as $P(A | B)$ and is calculated using the formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Note:

1. $P(B|A) = \frac{P(B \cap A)}{P(A)}$, where $P(A) \neq 0$.
2. In general, $P(A|B) \neq P(B|A)$.
3. $P(A|B)$ can be considered as the probability of occurrence of event $A \cap B$ with respect to the sample space B i.e. reduced sample space.

Example (part of **AJC Prelim 9740/2008/02/Q7**):

3 machines A, B and C produce 25%, 35% and 40% respectively of the golf balls manufactured by a factory. These balls are either yellow or white. Of the balls produced by A and B , 20% and 30% respectively are yellow. It is known that the probability of picking a yellow ball is 0.355. If 3 balls are picked randomly, find the probability that at least 1 is yellow given that all the balls picked are from machine A .

Using reduced sample space, $P(\text{at least 1 is yellow} | \text{all from } A) = 1 - 0.8^3 = 0.488$

Independent Events

Two events A and B are said to be independent if the occurrence of A does not affect that of B and vice versa.

To prove the independence of events A and B , we can show either one of the following

1. $P(A|B) = P(A)$
2. $P(B|A) = P(B)$
3. $P(A \cap B) = P(A)P(B)$

It can be easily shown that if A and B are independent, then the following pairs are also independent A and B' , A' and B' , A' and B .

We DO NOT assume independence unless we have proven it or condition is given in the question.

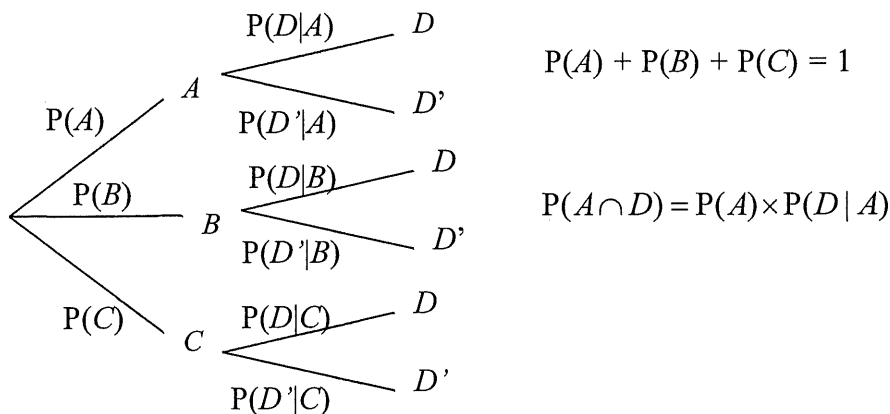
Common mistakes:

Mixed up the tests of “mutually exclusive” and “independence”.

Taking $P(A \cap B) = P(A)P(B)$ even when A and B are NOT independent.

Techniques in calculating probabilities

1. Systematic listing – example 9758/2019/P2/Q8iii.
2. Venn diagram – example 9758/2018/P2/Q7. This question involves 3 sets.
3. Tree diagram – though a useful technique, many students are too lazy to draw the diagram and hence made careless mistakes.



4. P&C – do not be confused when applying this technique. The counting for the denominator has to be consistent with that of the numerator. If order matters for the denominator, then it has to matter for the numerator.

Reminder: Probability could easily be embedded in another Statistics question due to its nature. Read each question carefully.

Revision Tutorial Questions**Source: YJC/2018/JC2/CT1/Q1**

- 1 3 families went to a cinema to watch a movie together. Each family consists of a father, mother, child and domestic helper. The surnames of the three families are Chua, Lee and Wong. The 12 people are seated in a row of 12 seats. Find the number of arrangements if
- (i) the children are seated next to each other, [1]
 - (ii) the 4 members in each family are seated next to each other, [2]
 - (iii) each domestic helper is seated next to the child in the family, and each family is seated together, [2]
 - (iv) all three mothers are next to each other, all four members of the Chua family are next to each other and all four members of the Wong family are next to each other. [3]
- After the movie, the 3 families proceed for a dinner at a round table with 12 seats.
- (v) In how many ways can the 12 people sit so that no two children are next to each other? [2]

Source: DHS/2017/Prelim/02/Q10(a)

- 2 The word DISTRIBUTION has 12 letters.
- (i) Find the number of different arrangements of the 12 letters that can be made. [1]
 - (ii) Find the number of different arrangements which can be made if there are exactly 8 letters between the two Ts. [3]
- One of the I's is removed from the word and the remaining letters are arranged randomly.
- (iii) Find the probability that no adjacent letters are the same. [4]

Source: AJC/2018/JC2/CT/02/Q2

- 3 A student has two routes to get to school. The probability that he chooses the first route on any day is p where $0 < p < 1$, and the probability of him being late for school is 0.1 if he chooses the first route and 0.2 for the second route. There is a probability of 0.14 that a student is late for school on any given day.
- (i) Find the value of p . [2]
 - (ii) Given that he is not late for school, find the probability that he chose the first route. [2]
 - (iii) Find the probability that the student is late for school more than once in three days. [2]

Source: HCI/2018/JC2/CT1/Q9

- 4 (a) In a particular college, students must choose 4 subjects to read. Subjects are classified as either 'Arts' or 'Science'. The available 'Arts' subjects are Economics, Geography, History and Literature, and the available 'Science' subjects are Biology, Chemistry, Mathematics and Physics. Each student must read at least one 'Arts' subject and at least one 'Science' subject.
- (i) Find the number of ways that a student can choose his 4 subjects. [2]
- The subjects are further classified into 2 levels, H1 and H2. Each student must read 3 subjects at H2 level and 1 subject at H1 level, and he cannot read the same subject at both H1 and H2 levels.
- (ii) Find the number of ways that a student can choose his 3 H2 and 1 H1 subjects. [1]
- In addition to the rules above, a student who reads H2 Biology, H2 Chemistry or H2 Physics must also read H2 Mathematics.
- (iii) Find the number of ways that a student can choose his 3 H2 and 1 H1 subjects with this additional rule. [3]
- (b) It is known that among the students in a college, 5% read Geography and 8% read History. There is a 20% chance that a student reads Geography given that he reads History.
- (i) Show that the probability that a randomly chosen student reads both Geography and History is 0.016. [1]
- (ii) Find the probability that a randomly chosen student reads either Geography or History but not both. [2]
- (iii) State, with a reason, whether the events 'a student reads Geography' and 'a student reads History' are independent. [1]

Source: ACJC/2018/JC2/CT1/Q3 (modified)

- 5 Given that $P(B \cup C) = \frac{3}{4}$ and $P(B' \cap C) = \frac{1}{5}$,
- (i) find $P(C)$ if B and C are two mutually exclusive events, [1]
- (ii) find $P(C)$ if B and C are two independent events. [4]

Source: IJC/2017/Prelim/02/Q6

- 6 Seven red counters and two blue counters are placed in a bag. All the counters are indistinguishable except for their colours. Clark and Kara take turns to draw a counter from the bag at random with replacement. The first player to draw a blue counter wins the game and the game ends immediately.

If Clark draws first, find the probability that

- (i) Clark wins the game at his third draw, [2]
 (ii) Kara wins the game. [3]

Source: RVHS/2018/JC2/CT/Q7

- 7 The numbers on the faces of a fair tetrahedron blue die are 1, 2, 3 and 4; while the numbers on the faces of another fair tetrahedron red die are -1 , 0, 1 and 2. Both dice are rolled once and the numbers on the faces in contact with the table are noted.

The random variable X denotes the sum of the numbers obtained from the 2 dice.

		Blue Die				
		+	1	2	3	4
Red Die	-1					
	0					
	1					
	2					

- (i) By completing the above table which shows the possible values of X , construct the probability distribution table for X . [2]
 (ii) Find $\text{Var}(X)$. [2]

The following events A and B are defined as follows:

A : X is an odd number

B : X is more than 2

- (iii) Find $P(A|B)$ and determine if A and B are independent. [3]